IBM APPLIED DATA SCIENCE CAPSTONE

Predicting Falcon 9 First-Stage Landings

Data collection, exploration, interactive analytics and machine learning to estimate whether a booster will land — and what a launch is likely to cost.

Duong October 2026

OUTLINE

What this deck covers

01 Executive Summary

02 Introduction

03 Methodology

04 Results Overview

05 Insights from EDA (Visual + SQL)

06 Launch Site Proximity Analysis

07 Dashboard with Plotly Dash

08 Predictive Analysis

09 Conclusion

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EXECUTIVE SUMMARY

Landing outcome is predictable at 83% test accuracy

Methodology

- Collected launch data via SpaceX REST API and Wikipedia web scraping
- Cleaned 90 Falcon 9 launches; built a binary landing label
- Explored with Seaborn charts and 10 SQL queries
- Mapped launch sites in Folium; built a Plotly Dash app
- Tuned LR, SVM, Decision Tree and KNN with 10-fold GridSearchCV

Results

- Overall landing success rate: 66.7% (60 of 90)
- Success rose from 0% (2010–2013) to 90% in 2019
- ES-L1, GEO, HEO and SSO orbits: 100% success; GTO: ~52%
- LR, SVM and KNN tie at 83.3% test accuracy
- Main error: false positives (failed landings predicted as landed)

INTRODUCTION

Reusing the first stage is what makes Falcon 9 cheap

Background

SpaceX advertises Falcon 9 launches at about $62 million, versus upward of $165 million from other providers. Much of the saving comes from landing and reusing the first stage.

If we can predict whether the first stage will land, we can estimate the cost of a launch — useful for a competitor bidding against SpaceX.

Questions to answer

- Which factors — payload, orbit, launch site, flight history — drive a successful landing?
- How have landing outcomes changed over time?
- Can a classifier predict landing success reliably enough to price a launch?

03

Methodology

From raw API and web data to a tuned classifier, in six steps.

METHODOLOGY

Executive summary of the methodology

1 · Data collection

SpaceX REST API (v4) + Wikipedia "List of Falcon 9 launches" scraped with BeautifulSoup

2 · Data wrangling

Filtered to Falcon 9, filled missing payloads, mapped 8 landing outcomes to a 0/1 Class label

3 · EDA

Seaborn/Matplotlib charts and SQL queries on an SQLite table

4 · Interactive maps

Folium markers, clusters and distance lines around each launch site

5 · Dashboard

Plotly Dash app with site dropdown, pie chart and payload slider

6 · Classification

Standardize, split 80/20, tune 4 models with GridSearchCV (cv=10)

METHODOLOGY · DATA COLLECTION

Collecting launch records from the SpaceX REST API

1. GET /v4/launches/past (static JSON snapshot)
2. Normalize JSON with pd.json_normalize
3. Resolve IDs: rocket, launchpad, payload, core
4. Build launch_dict → DataFrame
5. Keep Falcon 9 only; fill PayloadMass with mean
6. Export dataset_part_1.csv

Key fields captured

FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial, Longitude, Latitude

Notebook

[GitHub URL — data-collection-api.ipynb]

METHODOLOGY · DATA COLLECTION

Scraping launch history from Wikipedia

1. GET the Falcon 9 launch page (HTTP 200)
2. Parse HTML with BeautifulSoup
3. find_all('table') → 29 tables found
4. Extract column names from the header row
5. Loop rows, fill launch_dict per column
6. DataFrame → spacex_web_scraped.csv

Columns extracted

Flight No., Date and time, Launch site, Payload, Payload mass, Orbit, Customer, Launch outcome — plus Version Booster and Booster landing

Notebook

[GitHub URL — webscraping.ipynb]

METHODOLOGY · DATA WRANGLING

Turning eight landing outcomes into one label

1. Check missing values (LandingPad 28.9% null)
2. Count launches per site and per orbit
3. Count each landing outcome (8 categories)
4. Define bad_outcomes (5 failure / no-attempt types)
5. Class = 0 if bad outcome, else 1
6. Export dataset_part_2.csv (90 rows)

What the counts showed

Sites: CCAFS SLC 40 = 55, KSC LC 39A = 22, VAFB SLC 4E = 13

Top orbits: GTO 27, ISS 21, VLEO 14

Outcomes: True ASDS 41, None None 19, True RTLS 14

Mean of Class = 0.667

Notebook: [GitHub URL — data-wrangling.ipynb]

METHODOLOGY · EDA WITH VISUALIZATION

Each chart answers one question

Chart	Type	Why
Flight Number vs Payload	Scatter	Does experience or payload drive landing?
Flight Number vs Launch Site	Scatter	Do sites differ as launches accumulate?
Payload vs Launch Site	Scatter	Are heavy payloads tied to specific sites?
Success rate by Orbit	Bar	Which orbits land most reliably?
Flight Number vs Orbit	Scatter	Does orbit success improve over time?
Payload vs Orbit	Scatter	How does payload interact with orbit?
Yearly success trend	Line	How has the success rate evolved?

Then one-hot encoded Orbit, LaunchSite, LandingPad, Serial → 80 numeric features (dataset_part_3.csv) · [GitHub URL — eda-dataviz.ipynb]

METHODOLOGY · EDA WITH SQL

Ten SQL queries on an SQLite table

1 Unique launch sites

2 5 sites beginning with 'CCA'

3 Total payload for NASA (CRS)

4 Average payload of F9 v1.1

5 First successful ground-pad landing

6 Drone-ship boosters, 4000–6000 kg

7 Mission outcomes by type

8 Boosters with maximum payload

9 2015 drone-ship failures by month

10 Landing outcomes ranked, 2010–2017

Spacex.csv loaded with pandas → SPACEXTABLE (rows with null Date removed), queried via %sql magic · [GitHub URL — eda-sql.ipynb]

METHODOLOGY · INTERACTIVE MAP WITH FOLIUM

Map objects and why they were added

Circles + text markers

Highlight each launch site's coordinates and label it by name.

MarkerCluster

Group the many launches at one site; green = landed, red = failed.

MousePosition

Read coordinates of nearby coastline, rail, highway and city.

PolyLines + distance labels

Measure the distance from a site to each proximity feature.

Notebook: [GitHub URL — launch-site-location-folium.ipynb]

METHODOLOGY · DASHBOARD WITH PLOTLY DASH

Two linked views, two controls

Launch site dropdown

Choose All Sites or one site; drives both charts via callbacks.

Success pie chart

All sites: share of successes by site. One site: success vs failure.

Payload range slider

Filter launches between 0 and 10,000 kg.

Payload vs outcome scatter

Shows how success correlates with payload, colored by booster version.

Code: [GitHub URL — spacex_dash_app.py]

METHODOLOGY · PREDICTIVE ANALYSIS

Building, tuning and evaluating four classifiers

1. Y = Class; X = 80 one-hot features
2. StandardScaler on X
3. train_test_split 80/20 → 72 train, 18 test
4. GridSearchCV (cv=10) per model
5. Score on test set + confusion matrix
6. Compare CV and test accuracy; pick model

Models tuned

Logistic Regression — C, penalty, solver

SVM — kernel, C, gamma

Decision Tree — criterion, splitter, depth, features, leaf/split size

KNN — n_neighbors, algorithm, p

Notebook: [GitHub URL — ml-prediction.ipynb]

04 · RESULTS OVERVIEW

Three numbers that frame the findings

66.7%

EDA: overall landing success across 90 Falcon 9 launches

90%

EDA trend: peak yearly success rate, reached in 2019 (from 0% in 2010–2013)

83.3%

Predictive analysis: best test accuracy (LR, SVM, KNN)

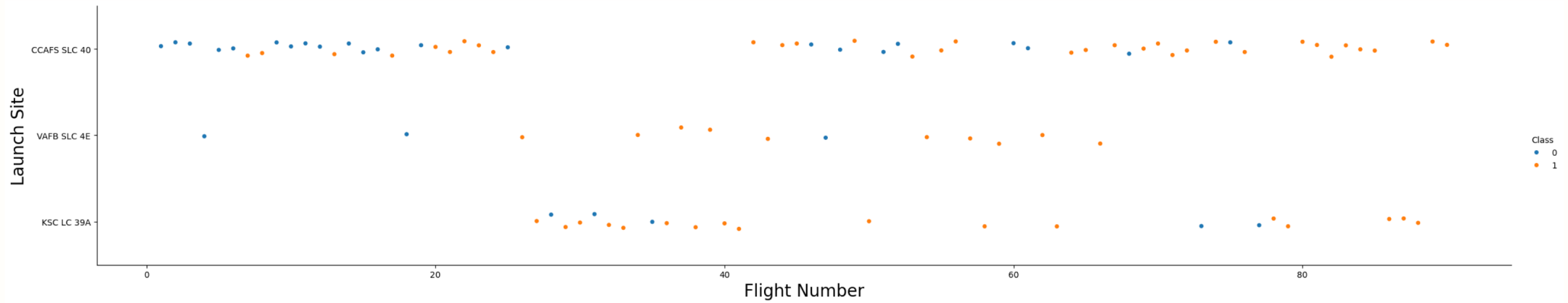
05

Insights drawn from EDA

Six charts on launch site, payload, orbit and time — then ten SQL queries.

EDA · FLIGHT NUMBER VS LAUNCH SITE

Later flights land more often at every site



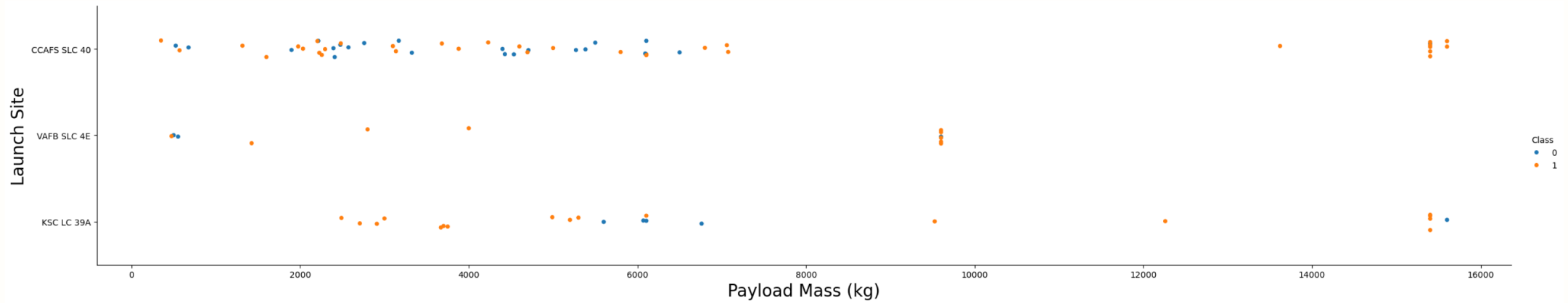
CCAFS SLC 40 hosts most launches; its early flights (1–25) were mostly failures (blue).

KSC LC 39A and **VAFB SLC 4E** came later and show high success from the start.

After flight ~40, successes (orange) dominate at all sites — experience matters.

EDA · PAYLOAD VS LAUNCH SITE

Heavy payloads launch from Florida and mostly land



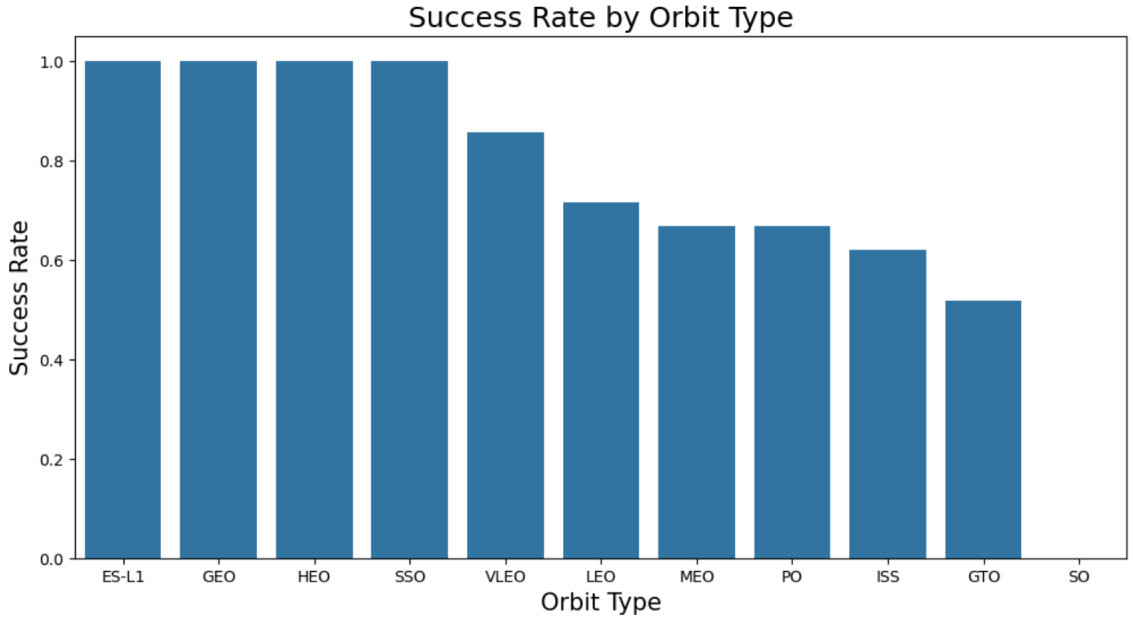
VAFB SLC 4E has no launches above 10,000 kg.

Above ~13,000 kg (CCAFS, KSC) almost every landing succeeded.

The 2,000–7,000 kg band at CCAFS is mixed — payload alone does not decide the outcome.

EDA · SUCCESS RATE BY ORBIT TYPE

Four orbits land every time; GTO barely half



100%

ES-L1, GEO, HEO, SSO — note ES-L1, GEO, HEO have just 1 launch each

~86% · ~71%

VLEO (14 launches) and LEO (7)

~52% – 67%

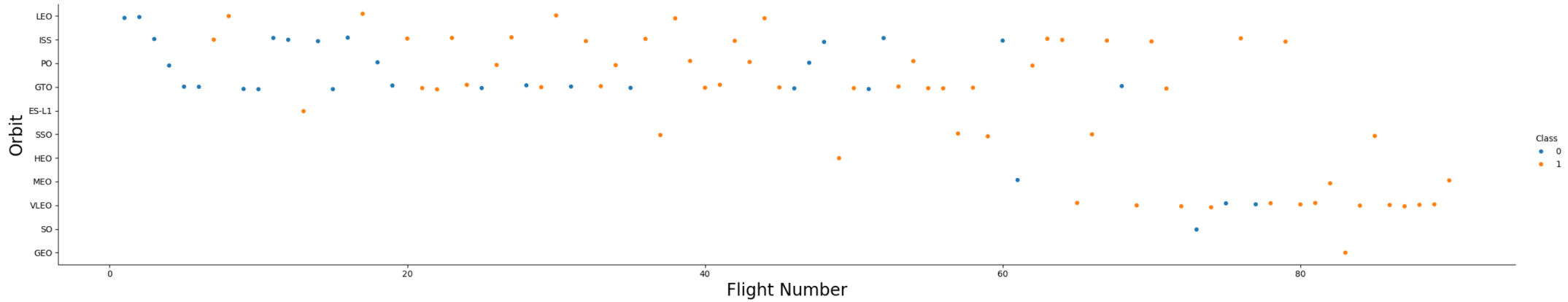
MEO, PO, ISS and GTO — GTO is the weakest high-volume orbit (27 launches)

0%

SO — a single failed launch

EDA · FLIGHT NUMBER VS ORBIT TYPE

LEO improves with experience; GTO does not



In **LEO**, success follows flight number: early flights failed, later ones landed.

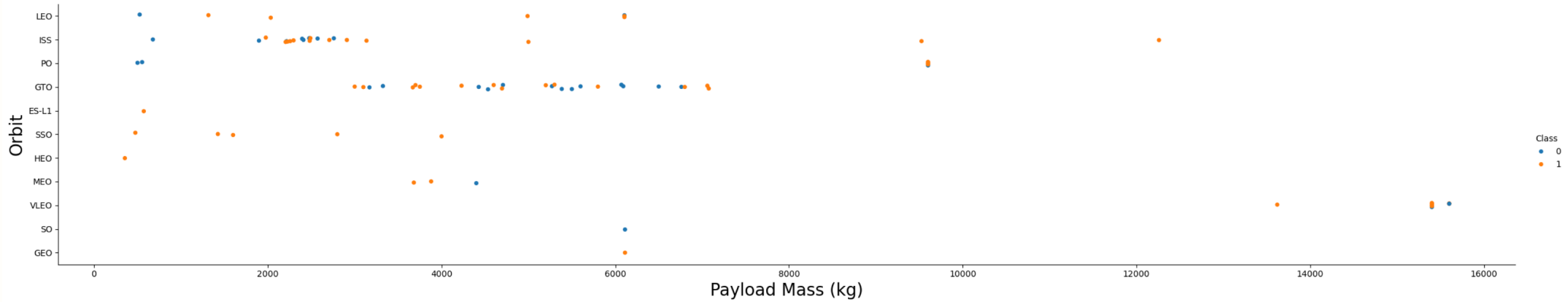
In **GTO**, outcomes stay mixed across the whole range of flight numbers.

VLEO appears only after flight ~65 and is mostly successful.

Falcon 9 Landing Prediction · EDA · Class 1 = landed (orange), 0 = not landed (blue)

EDA · PAYLOAD VS ORBIT TYPE

Heavy payloads help in LEO, ISS and PO — not in GTO



For **PO, LEO and ISS**, heavier payloads show more successful landings.

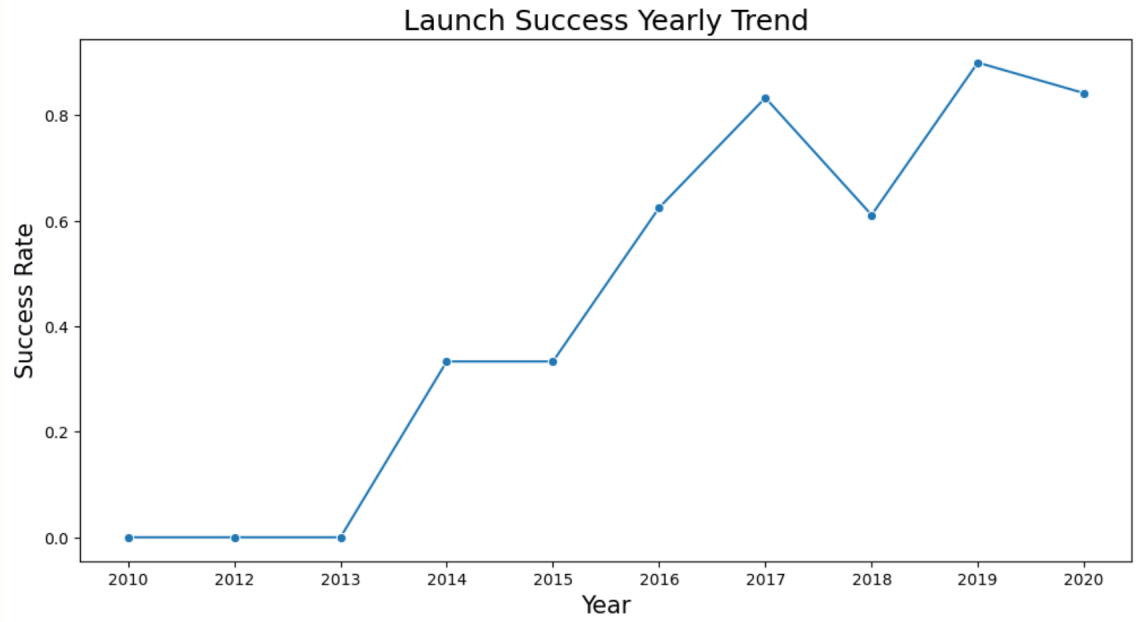
For **GTO** (3,000–7,000 kg) successes and failures overlap — hard to separate.

VLEO carries the heaviest loads (~13,600–15,600 kg).

Falcon 9 Landing Prediction · EDA · Class 1 = landed (orange), 0 = not landed (blue)

EDA · LAUNCH SUCCESS YEARLY TREND

From zero to 90% in six years



2010 – 2013

0% — no successful landings

2014 – 2017

Steady climb: ~33% → ~83%

2018

Dip to ~61%

2019 – 2020

Peak ~90%, then ~84%

EDA WITH SQL · QUERY 1

All launch site names

Query

```
SELECT DISTINCT "Launch_Site"
FROM SPACEXTABLE;
```

Result

```
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40
```

DISTINCT removes duplicates: four site names, three in Florida and one in California. CCAFS LC-40 and SLC-40 are the same pad under old and new names.

EDA WITH SQL · QUERY 2

Launch sites beginning with ’CCA’

Query

```
SELECT *
FROM SPACEXTABLE
WHERE "Launch_Site"
LIKE 'CCA%'
LIMIT 5;
```

Date	Booster	Payload kg	Landing
2010-06-04	F9 v1.0 B0003	0	Failure (parachute)
2010-12-08	F9 v1.0 B0004	0	Failure (parachute)
2012-05-22	F9 v1.0 B0005	525	No attempt
2012-10-08	F9 v1.0 B0006	500	No attempt
2013-03-01	F9 v1.0 B0007	677	No attempt

All five are early CCAFS LC-40 flights to LEO with light payloads — none recovered the booster.

EDA WITH SQL · QUERY 3

Total payload mass carried for NASA (CRS)

Query

```
SELECT SUM("PAYLOAD_MASS__KG_")
AS Total_Payload_Mass_KG
FROM SPACEXTABLE
WHERE "Customer" = 'NASA (CRS)';
```

Result

45,596 kg

SUM aggregates every Commercial Resupply Services mission to the ISS — NASA is a major, recurring customer.

EDA WITH SQL · QUERY 4

Average payload mass carried by F9 v1.1

Query

```
SELECT AVG("PAYLOAD_MASS__KG_")
AS Avg_Payload_Mass_KG
FROM SPACEXTABLE
WHERE "Booster_Version" = 'F9 v1.1';
```

Result

2,928.4 kg

The early v1.1 booster carried modest loads — far below the 15,600 kg maximum later reached by Block 5.

EDA WITH SQL · QUERY 5

First successful ground-pad landing

Query

```
SELECT MIN("Date")
AS First_Successful_Ground_Landing
FROM SPACEXTABLE
WHERE "Landing_Outcome" =
'Success (ground pad)';
```

Result

2015-12-22

The first ground-pad recovery came five years after the first Falcon 9 launch — the turning point visible in the yearly trend.

EDA WITH SQL · QUERY 6

Drone-ship successes with 4,000–6,000 kg payload

Query

```
SELECT "Booster_Version"
FROM SPACEXTABLE
WHERE "Landing_Outcome" =
'Success (drone ship)'
AND "PAYLOAD_MASS__KG_" > 4000
AND "PAYLOAD_MASS__KG_" < 6000;
```

Result

```
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2
```

All four are Full Thrust boosters; two (.2 suffix) were already on their second flight — reuse was working.

EDA WITH SQL · QUERY 7

Total successful and failed mission outcomes

Query

```
SELECT "Mission_Outcome",
COUNT(*) AS Total
FROM SPACEXTABLE
GROUP BY "Mission_Outcome";
```

Mission outcome	Total
Success	98
Success (trailing space)	1
Success (payload status unclear)	1
Failure (in flight)	1

100 of 101 missions succeeded. Mission success ≠ landing success — the landing label is what we predict.

EDA WITH SQL · QUERY 8

Boosters that carried the maximum payload

Query (subquery)

```
SELECT "Booster_Version"
FROM SPACEXTABLE
WHERE "PAYLOAD_MASS__KG_" = (
SELECT MAX("PAYLOAD_MASS__KG_")
FROM SPACEXTABLE
);
```

B1048.4	B1049.4	B1051.3
B1056.4	B1048.5	B1051.4
B1049.5	B1060.2	B1058.3
B1051.6	B1060.3	B1049.7

All 12 are F9 B5 (Block 5) boosters, on their 2nd to 7th flight — the heaviest missions use the most-reused hardware.

EDA WITH SQL · QUERY 9

2015 drone-ship landing failures by month

Query

```
SELECT substr("Date",6,2) AS Month,
"Landing_Outcome",
"Booster_Version", "Launch_Site"
FROM SPACEXTABLE
WHERE substr("Date",0,5) = '2015'
AND "Landing_Outcome" =
'Failure (drone ship)';
```

Month	Booster	Site
01	F9 v1.1 B1012	CCAFS LC-40
04	F9 v1.1 B1015	CCAFS LC-40

Both 2015 drone-ship failures used F9 v1.1 boosters from CCAFS LC-40 — early attempts before the Full Thrust upgrade.

EDA WITH SQL · QUERY 10

Landing outcomes ranked, 2010–2017

Query

```
SELECT "Landing_Outcome",
COUNT(*) AS Outcome_Count
FROM SPACEXTABLE
WHERE "Date" BETWEEN
'2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome"
ORDER BY Outcome_Count DESC;
```

Landing outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Only 8 of 31 launches recovered the booster (5 drone ship, 3 ground pad); 10 made no attempt.

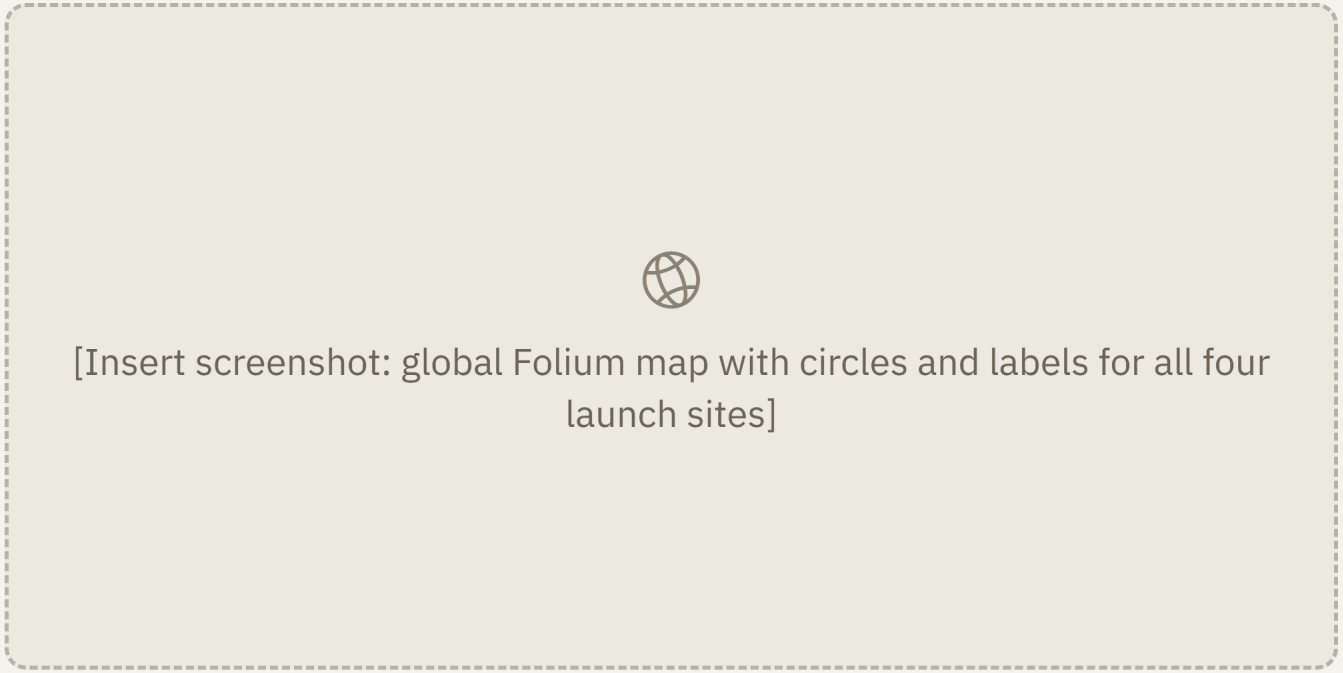
06

Launch Site Proximity Analysis

Where the sites are, how each performs, and what lies around them.

FOLIUM · ALL LAUNCH SITES

Every site sits on a US coastline



Three sites on Florida's Atlantic coast (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A), one on California's Pacific coast (VAFB SLC-4E).

Florida (~28.5°N) is the US territory closest to the equator, giving extra eastward velocity for GTO launches.

Coastal locations let rockets fly over water, away from people.

FOLIUM · LAUNCH OUTCOMES PER SITE

Color-coded outcomes at each site

[Insert screenshot: zoomed MarkerCluster at one site, green = landed, red = failed]

Each launch is a marker inside a cluster; green for Class 1, red for Class 0.

[Your observation, e.g. which site shows the highest share of green markers]

[Your observation on CCAFS LC-40's early red markers]

FOLIUM · PROXIMITIES

Close to coast and rail, far from cities



Nearest	Distance
Coastline	[__ km]
Railway	[__ km]
Highway	[__ km]
City	[__ km]

Sites stay near transport for logistics but keep a safety buffer from populated areas.

07

Dashboard with Plotly Dash

Interactive views of success by site and by payload range.

DASHBOARD · SUCCESS BY SITE

Share of successful launches across all sites



[Insert screenshot: pie chart, dropdown = All Sites]

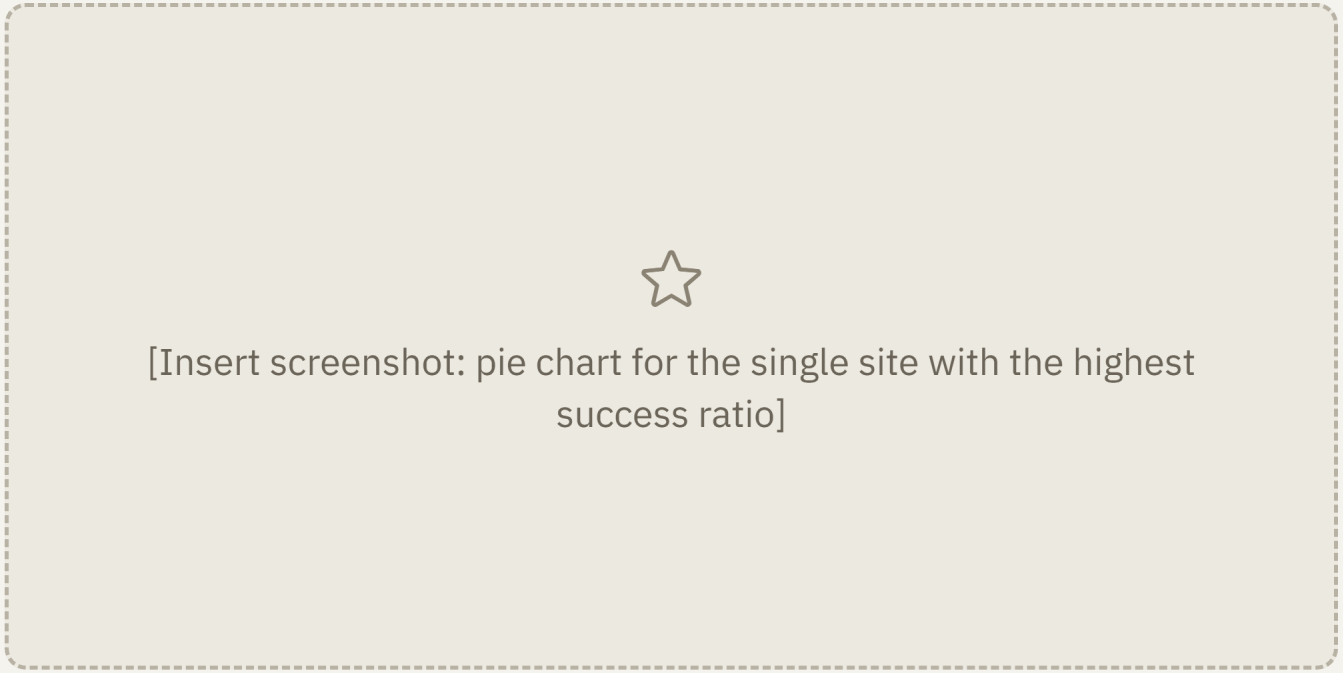
Each slice = one site's share of all successful landings.

[Site with the largest slice and its %]

[Site with the smallest slice and its %]

DASHBOARD · BEST SITE

The site with the highest success ratio



[Site name]: [__]% success, [__]% failure.

[One-line reason, e.g. it opened later, after landing technology had matured]

DASHBOARD · PAYLOAD VS OUTCOME

Success across payload ranges and booster versions



[Payload range with the highest success rate]

[Payload range with the lowest success rate]

[Booster version with the highest success rate, e.g. FT]

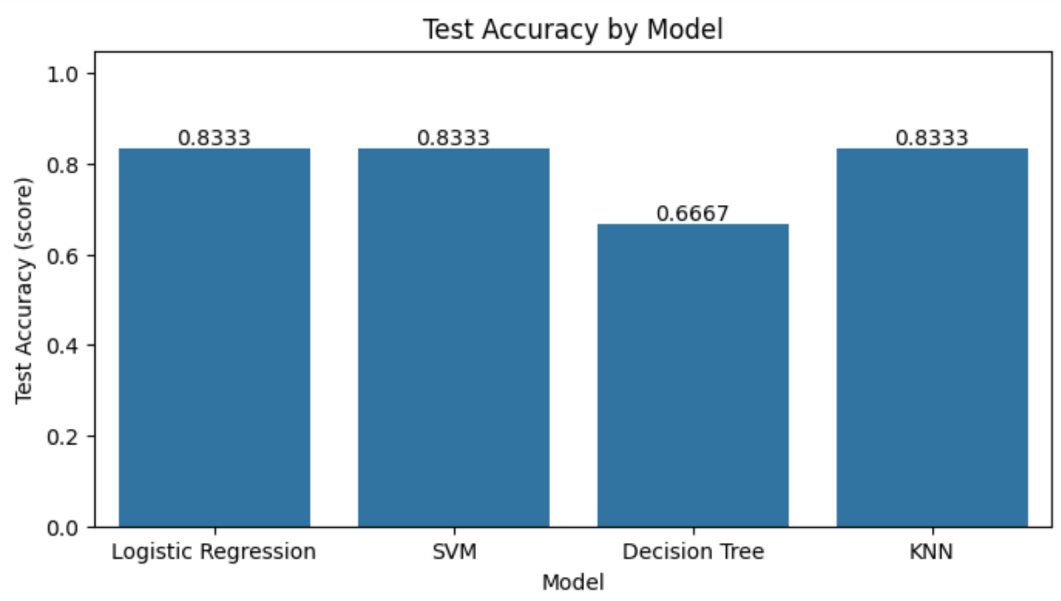
08

Predictive Analysis

Four classifiers, tuned with 10-fold cross-validation, scored on 18 held-out launches.

CLASSIFICATION ACCURACY

Three models tie on test data; the tree overfits



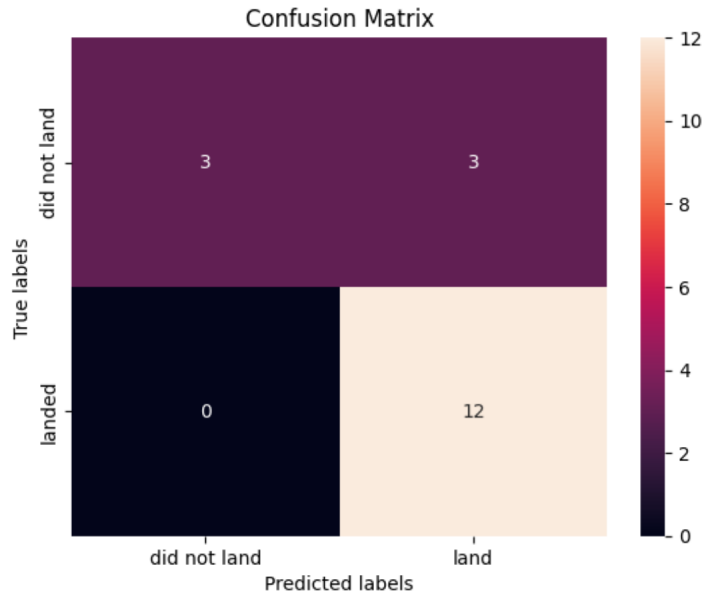
Model	CV	Test
Log. Regression	0.846	0.833
SVM	0.848	0.833
Decision Tree	0.875	0.667
KNN	0.848	0.833

The Decision Tree has the best CV score but the worst test score — a 21-point drop that signals overfitting.

Pick: Logistic Regression — top test accuracy, stable CV, simplest to explain.

CONFUSION MATRIX · LOGISTIC REGRESSION

No missed landings — but three false alarms



True positive
12

True negative
3

False positive
3

False negative
0

Every landed booster was predicted correctly (recall = 100%). Half of the 6 failed landings were predicted as landed — the model is optimistic.

SVM and KNN produce the identical matrix.

09 · CONCLUSION

What the data says about landing a Falcon 9

Experience

Success rose from 0% (2010–2013) to 90% (2019); later flights land more often at every site.

Orbit

SSO, VLEO and LEO land reliably; GTO (~52%) is the hardest high-volume orbit.

Payload & site

Very heavy payloads (>13,000 kg) mostly landed; VAFB never flew above 10,000 kg.

Prediction

Logistic Regression reaches 83.3% test accuracy with zero missed landings — good enough for a first cost estimate.

Next: more launches and features (weather, booster reuse count) to cut false positives; re-test on a larger hold-out set.

APPENDIX · BEST HYPERPARAMETERS

GridSearchCV results (cv = 10)

Model	Best parameters	CV	Test
Logistic Regression	C = 0.01, penalty = l2, solver = lbfgs	0.8464	0.8333
SVM	kernel = sigmoid, C = 1.0, gamma = 0.0316	0.8482	0.8333
Decision Tree	gini, splitter = best, max_depth = 10, max_features = sqrt, min_samples_leaf = 4, min_samples_split = 10	0.8750	0.6667
KNN	n_neighbors = 10, algorithm = auto, p = 1	0.8482	0.8333

Data: 90 launches × 80 features after one-hot encoding; StandardScaler; train_test_split(test_size = 0.2, random_state = 2) → 72 train / 18 test.

APPENDIX · CODE AND DATA

Project repository

[https://github.com/your-username/your-repo]

File	Step
jupyter-labs-spacex-data-collection-api.ipynb	Data collection — API
jupyter-labs-webscraping.ipynb	Data collection — web scraping
labs-jupyter-spacex-Data wrangling.ipynb	Data wrangling
jupyter-labs-eda-sql-coursera_sqllite.ipynb	EDA with SQL
edadataviz.ipynb	EDA with visualization
[launch_site_location.ipynb]	Folium maps
[spacex_dash_app.py]	Plotly Dash
SpaceX_Machine Learning Prediction_Part_5.ipynb	Classification